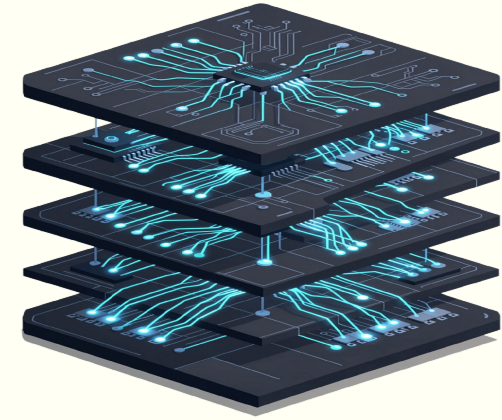


Undercomplete & Regularized Autoencoders



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INTRODUCTION

- Autoencoders are neural networks used to learn a compact representation of input data.
- They have two main parts: Encoder and Decoder.
- Undercomplete Autoencoder uses a hidden representation with fewer dimensions than the input.
- It learns the most important features while removing unnecessary information.

OBJECTIVES

- To understand the concept of Undercomplete Autoencoders.
- To study the architecture of Regularized Autoencoders.
- To understand the encoding and decoding process.
- To learn how autoencoders perform dimensionality reduction.
- To study the role of reconstruction loss.

Undercomplete Autoencoders

When the code dimension is strictly less than the input dimension, the autoencoder is forced to compress — learning to retain only the most salient features of the training data.

Compression Constraint

Code dim < input dim forces the network to discard redundancy and encode the most informative structure.

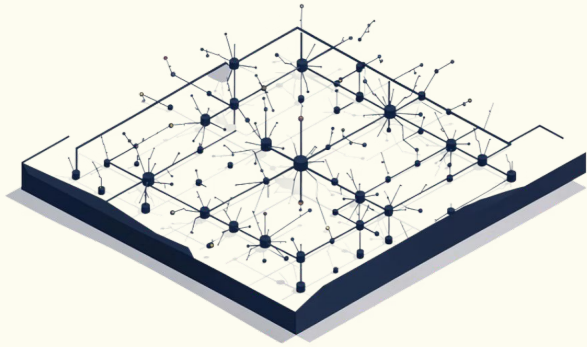
Linear Case $\equiv$ PCA

With a linear decoder and MSE loss, the undercomplete autoencoder learns to span exactly the same subspace as Principal Component Analysis.

Nonlinear Generalization

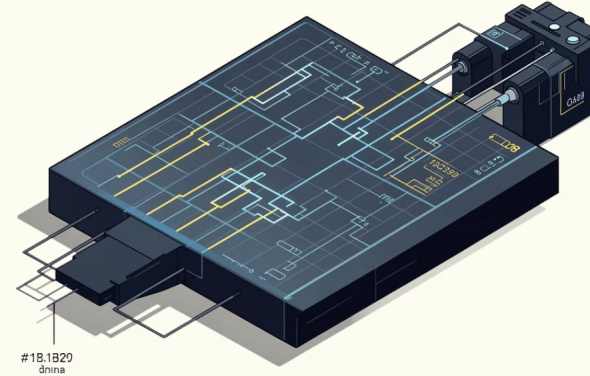
Nonlinear encoders/decoders extend PCA to curved manifolds — a strictly more powerful dimensionality reduction.

Sparse & Denoising Autoencoders



Sparse Autoencoder

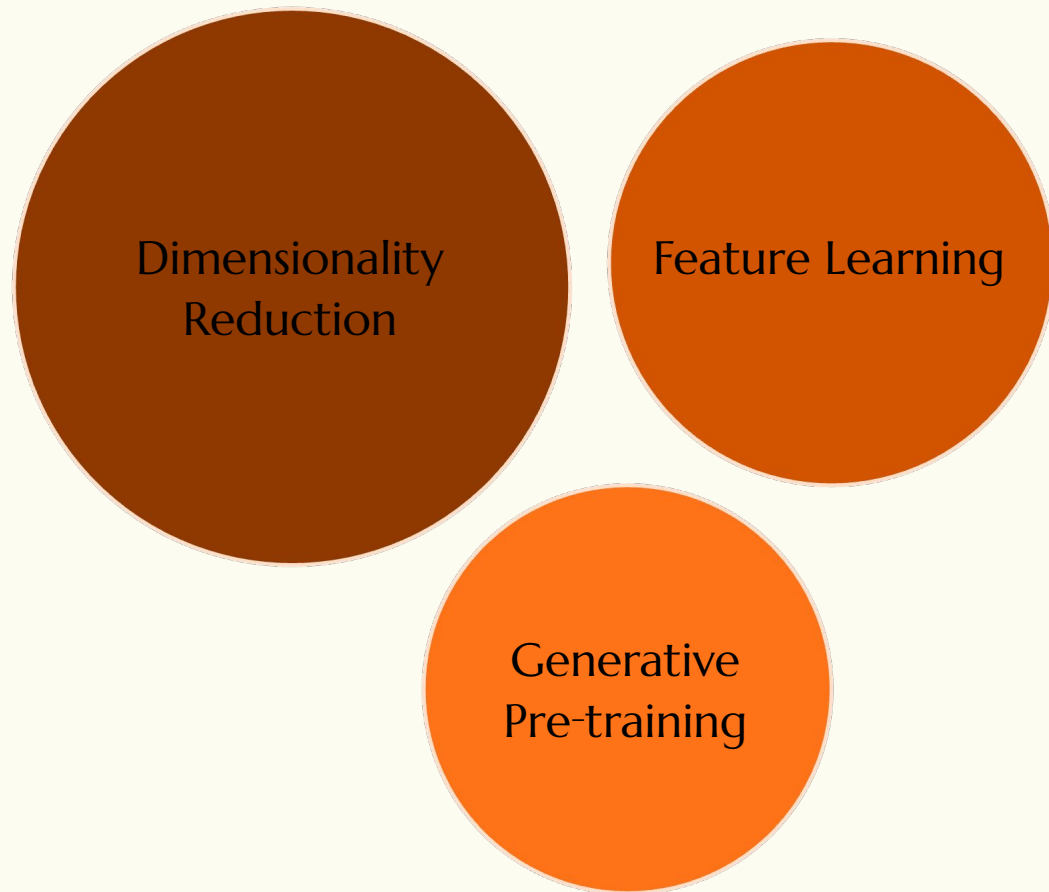
An L1 penalty on hidden activations encourages most units to remain **inactive** for any given input. Sparse distributed representations are not only computationally efficient — they reflect how real-world data is generated, isolating independent factors of variation across distinct units.



Denoising Autoencoder

The model receives a corrupted version of the input and must reconstruct the clean original. This corruption-reconstruction objective implicitly captures the structure of the data-generating density — the model must learn what is signal and what is noise, yielding robust and generalizable features.

Applications: Dimensionality Reduction & Feature Learning



From Compression to Generation

Autoencoders were originally developed for dimensionality reduction and feature learning. Theoretical connections to latent variable models have since brought them to the forefront of generative modeling.

- **Dimensionality reduction:** Encoder output is a compact, task-ready embedding
- **Feature learning:** Latent codes disentangle underlying factors of variation
- **Anomaly detection:** High reconstruction loss flags out-of-distribution inputs
- **Pre-training:** Unsupervised initialization for supervised downstream tasks

CONCLUSION

- Undercomplete Autoencoders learn compact representations by reducing the input dimensions.
- Regularized Autoencoders learn meaningful features by using regularization techniques.
- Both use encoder and decoder networks to reconstruct the input.
- Reconstruction loss helps measure how accurately the input is recovered.